attacker might send a request for /data/../../../etc/shadow.[12] If the webserver doesn't carefully check paths, then it may serve up the local password file.

If the attacker manages to steal the long-term secret key used by the device, then it can be impossible to regain security without human involvement. In the worst case, the device may need to be replaced or restored to factory settings and reconfigured. The ratcheting mechanisms discussed in section 12.4.3 do not protect against compromise, because if the attacker ever gains access to the current ratchet key, they can easily calculate all future keys.

Hardware security measures, such as a secure element, TPM, or TEE (see section 12.4.1) can provide post-compromise security by ensuring that an attacker never directly gains access to the secret key. An attacker that has active control of the device can use the hardware to compromise communications while they have access, but once that access is removed, they will no longer be able to decrypt or interfere with future communications.

A weaker form of post-compromise security can be achieved if an external source of key material is mixed into a ratcheting process periodically. If the client and server can agree on such key material without the attacker learning it, then any new derived keys will be unpredictable to the attacker and security will be restored. This is weaker than using secure hardware, because if the attacker has stolen the device's key then in principle they can eavesdrop or